

System Analysis & Design (SAD) with HCI Laboratory Activity

Course: Systems Analysis and Design / Human-Computer Interaction

Activity Title: Case Study & System Modeling: Enterprise Bank Next-Gen ATM Overhaul

Target Output: DFD Suite, ERD Schema, Low-Fidelity Prototype, and System Design Rationale

Format: Small Groups (3–4 Members) | **Total Points:** 100 Points

1. Business Scenario & Problem Context

Enterprise Bank operates a nationwide network of 2,500 Automated Teller Machines (ATMs). The bank's executive board has mandated a full architectural overhaul of their current system due to severe operational, security, and usability challenges:

- **Legacy Hardware & Security Risks:** Existing machines rely on physical magnetic stripe/EMV cards and 4-digit PIN keypads. Skimming incidents, card-trapping, and shoulder-surfing PIN theft have increased by 35% over the past year.
- **High Transaction Latency:** Navigating rigid 6-screen menu trees takes an average of 45 seconds per transaction, leading to long queues during peak payroll periods.
- **Reactive Cash & Maintenance Operations:** Branch operators only dispatch cash replenishment or service technicians *after* an ATM goes completely out of cash or reports an operational hardware failure.
- **Accessibility Barriers:** Fixed monitor angles and physical button placements fail compliance checks for wheelchair-bound and visually impaired customers.

Project Mandate: "Enterprise-Kiosk Vanguard"

Your team has been hired as System Analysts and HCI Engineers to design a modern, intelligent, and secure next-generation ATM system. The new system must integrate:

1. **Touchless Biometric Authentication:** Zero-card login using Near-Infrared (NIR) Palm Vein scanning or 3D Facial Liveness detection.
2. **Predictive AI Personalization:** Automatically rendering a 1-tap personalized dashboard based on user transaction habits upon login.
3. **Mid-Air Spatial Touch & Smart Privacy Glass:** Touchless mid-air gesture control paired with dynamic electrochromic glass that auto-fogs side viewing angles when a user is present.

4. **Predictive Cash & Telemetry Analytics:** Continuous automated monitoring of cash cassette levels and sensor health metrics to trigger automated maintenance and cash delivery dispatches before outages occur.

2. Student Tasks & Required Deliverables

Your team must complete and submit the following four (4) core deliverables:

Task 1: Data Flow Diagrams (DFD) — Process Modeling

Synthesize the system boundaries, external entities, inputs, outputs, and internal processes for the proposed system.

1. Context Diagram (Level 0 DFD):

- Draw the overall system boundary for the *Vanguard ATM System*.
- Identify all external entities (e.g., *Customer/User*, *Core Banking System*, *Fleet Operations Center*).
- Map all high-level data streams entering and leaving the system.

2. Level 1 DFD:

- Decompose the main system into at least four (4) key sub-processes:
 - 1.0 Authenticate & Contextualize User
 - 2.0 Process Smart Transaction
 - 3.0 Manage AI Cash & Hardware Telemetry
 - 4.0 Execute Emergency Lockdown & Privacy
- Include all necessary data stores (e.g., D1: Biometric Store, D2: Account Ledger, D3: Terminal Telemetry, D4: Cassette Inventory).

Task 2: Entity-Relationship Diagram (ERD) — Data Architecture

Design a logical database schema capable of supporting real-time biometric verification, transaction processing, predictive maintenance, and multi-cassette inventory tracking.

1. Construct a logical ERD showing entities, primary keys (PK), foreign keys (FK), and non-key attributes.
2. Ensure your diagram explicitly includes at least these 6 entities:
 - CUSTOMER, BIOMETRIC_PROFILE, ACCOUNT, TRANSACTION, ATM_TERMINAL, and CASSETTE_INVENTORY.

3. Specify exact cardinalities (1:1, 1:N, N:M) and state all business rules (e.g., *One Customer can enroll multiple Biometric Profiles; One ATM Terminal houses multiple Cassettes*).

Task 3: Low-Fidelity (Low-Fi) HCI Physical & Software Prototype

Using physical low-cost materials (cardboard, paper, sticky notes, markers, clear sheets), build a physical representation of the terminal and software interface.

1. Cardboard Physical Shell (Hardware Ergonomics):

- Build a 1:1 scale physical front panel of the kiosk showing screen placement angle, palm sensor resting zone, cash dispenser, and electrochromic privacy glass border.

2. Paper UI Screen Deck (Software Navigation):

- Draft paper screens representing the core user journey:
 - *Screen A:* Idle / Touchless Approach & Privacy Activation
 - *Screen B:* Biometric Scan & Verification
 - *Screen C:* Predictive 1-Tap Personalized Dashboard
 - *Screen D:* Real-time Dispense & E-Receipt Confirmation

3. Wizard of Oz Usability Simulation:

- Prepare your prototype to undergo a live 3-minute usability run where one team member acts as the "System Wizard" (manually sliding paper screens and sticky-note overlays in response to a test user's gestures).

Task 4: Systems Analysis & Design Rationale Report

Write a concise technical rationale (3–4 pages) addressing the following justification prompts:

1. **DFD Process Rationale:** Explain how your proposed DFD processes eliminate the latency and security vulnerabilities of the legacy system.
2. **ERD Structural Rationale:** Explain why BIOMETRIC_PROFILE is isolated from core CUSTOMER account data, and how tracking individual CASSETTE_INVENTORY rows prevents complete ATM lockouts.
3. **HCI Usability Justification:**
 - Cite specific usability principles (e.g., *Fitts's Law*, *Nielsen's Usability Heuristics*) used when positioning mid-air touch targets and predictive menu buttons.

- Explain why conducting Low-Fidelity paper/cardboard testing in the requirements phase reduces system development costs compared to testing coded software.

3. Submission Checklist

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	SUBMISSION CHECKLIST
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[]	1. Context Diagram (Level 0) & Level 1 DFD (Engineered / Cleanly Drawn)
[]	2. Complete ERD with Entities, PK/FKs, Attributes, & Cardinality Notations
[]	3. Physical Cardboard Kiosk Shell + Paper UI Screen Deck
[]	4. System Analysis & Design Rationale Written Report
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4. Evaluation & Grading Rubric

Assessment Criteria	Exemplary (90-100%)	Satisfactory (75-89%)	Needs Improvement (<75%)
Process Modeling (DFD) <i>(25 Points)</i>	Diagrams strictly adhere to Gane-Sarson / Yourdon notation. Level 0 and Level 1 are fully balanced with no orphan processes, black holes, or miracle inputs/outputs.	DFDs are complete, but minor balancing errors exist between Level 0 and Level 1, or notation style is inconsistent.	Missing required processes/data stores; invalid data flow connections present.
Data Modeling (ERD) <i>(25 Points)</i>	ERD correctly models all business requirements. Primary/foreign keys, attributes, and explicit cardinality relationships are accurate and normalized.	Entities and attributes are complete, but cardinalities are improperly labeled or normalization issues exist.	Missing key entities or relationship lines; primary and foreign keys are omitted.
HCI Low-Fi Prototyping	Physical cardboard shell and paper screen deck clearly demonstrate futuristic features (biometrics, spatial touch,	Prototype is functional but lacks physical ergonomic detail or dynamic screen	Prototype is incomplete, messy, or fails to model touchless/futuristic features.

Assessment Criteria	Exemplary (90-100%)	Satisfactory (75-89%)	Needs Improvement (<75%)
	privacy glass) and operate smoothly in WoZ testing.	overlays for key user actions.	
<i>(25 Points)</i>			
Design Rationale & Defense	Written report provides rigorous architectural justifications for DFD/ERD decisions and grounds HCI design in recognized usability heuristics.	Rationale is provided but relies on general opinions rather than formal SAD concepts or usability heuristics.	Missing rationale sections; fails to explain or defend proposed system choices.
<i>(25 Points)</i>			